ECG Classifier Projet Report

Group 17

1 Business/Application Understanding

1.1 Application Scenario

The electrocardiogram (ECG) is a medical test that records the electrical activity of the heart through electrodes placed on the skin. It is a standard, non-invasive tool for detecting heart diseases, which are the leading cause of death worldwide (World Health Organization, 2025).

Each heartbeat produces a characteristic waveform, changes in its shape or timing can be an indicator of early signs of heart diseases so it is important for doctors to identify these anomalies. To find them, a patient usually wears a Holter monitor that records the ECG for 24 to 48 hours. However, at around 70 beats per minute this is roughly 100,000 beats per day, which is far too many beats for a doctor to check them by hand. In recent years, health monitoring is moving from the clinic to the patient thanks to small wearable devices that can record the ECG continuously during daily life. The embedding of software that labels each beat into these small wearable devices can point doctors to the few beats that actually need a closer look.

The goal of this project is to develop a model able to classify heartbeats from an electrocardiogram and that is small enough to run on a wearable device instead of on a computer or a server. Running the model on the device (Edge-AI) itself has some clear advantages. The raw ECG does not have to leave the device, which is better for privacy. Sending a few labels uses much less radio power and data than streaming the full signal all day so the battery lasts longer. The device also keeps working when there is no internet connection.

The main users would be doctors and nurses who review the flagged beats, the patients who wear the monitor and companies that develop these health monitoring wearables.

Our project only covers the ECG part of such a system using data from the MIT-BIH Arrhythmia Database (Moody and Mark, 2005) available on PhysioNet (Pollard et al., 2026), so no motion-sensor data is used. It is a course project and not a medical device, and we do not claim that our model is clinically validated.

1.2 Data-Mining Objective

Input: for each heartbeat the model gets two things:

- The beat itself: a window of 128 samples from ECG lead MLII at 180 Hz (0.71 s), with 48 samples before the R-peak and 80 samples after it.
- Three RR intervals: the time since the previous beat, the time until the next beat, and the median of the last four gaps, which shows the patient's recent normal rhythm.

We include the RR intervals because an S beat often looks almost the same as a normal beat. The main difference is that it comes too early, which is easy to see when the time since the previous beat is compared with the recent median.

Output: one of the four classes in Table **??**. These have been grouped in the same way as in de Chazal et al. (2004). The model gives a score for each class and the class with the highest score is the final prediction.

Class	Name	Description
N	Normal-type beat	Normal beats, plus bundle branch block beats and atrial or nodal escape beats
S	Supraventricular ectopic beat	Premature beats that start above the ventricles (in the atria or the AV node)
V	Ventricular ectopic beat	Beats that start in the ventricles, mostly premature ventricular contractions (the class also includes 106 ventricular escape beats)
F	Fusion beat	A ventricular beat and a normal beat that happen at the same time and merge

Type of task: supervised multiclass classification with four classes and one label per beat.

1.3 Project Objectives

Our main aim is to develop a compact model that can tell N, S, V and F beats apart in patients it has not seen before and that works on the edge device MAX78002.

Our technical objectives are:

- Classify each heartbeat as N, S, V or F.
- Generalize to new, unseen patients. We train and select models only on DS1 and test once on DS2, and we always split by recording, never by individual beats.
- Detect the rare classes S and F reasonably well, not only the majority class N.
- Be robust to real ECG noise from the MIT-BIH Noise Stress Test Database (Moody et al., 1984).
- Keep the model compact and use only layer types the MAX78002 supports, such as 1D convolution, pooling, ReLU and a small fully connected output layer.
- Quantize the model to INT8 with only a small loss in performance.
- Get the model through the Analog Devices ai8x-training and ai8x-synthesis tools and generate C code for the MAX78002.
- Compare it with a simple baseline, to check if the neural network is worth the extra size.
- Keep everything reproducible with one config file and a fixed random seed.

1.4 Success Criteria

We will not judge the model on accuracy alone because of the class imbalance of the dataset. On the DS2 test dataset, a model that always predicts N would already get about 89% accuracy. Instead we use macro-F1 (the average F1 score of the four classes) and recall for the rare classes. All targets are measured on DS2 with the final INT8 model unless the table says otherwise. Table 1 provides a summary of the evaluation criteria we will use in the project.

Table 1: Evaluation and success criteria

Area	Criterion
Predictive performance	Macro-F1 of at least 0.55, and better than the baseline
Minority classes	Recall of at least 0.60 for S and at least 0.80 for V. F is reported without a fixed target, because 362 of the 388 F beats in DS2 (93%) come from one patient (record 213).
Noise robustness	At 12 dB macro-F1 stays at least 90% of the clean result, and at 6 dB at least 75%. 0 dB is only reported.
Quantization	INT8 macro-F1 is at most 0.03 lower than FP32, and S and V recall drop by at most 5 percentage points.
Model complexity	INT8 weights of at most 2 MB. We aim for fewer than 100,000 parameters.
MAX78002 compatibility	Model accepted by the toolchain, quantization completed, synthesis completed and C files generated.

1.5 Constraints

Limited memory. The model must fit within 2 MB, so we use a small 1D CNN with a short input. Downsampling to 180 Hz helps since the 0.71 s window is 128 samples long instead of the 256 samples at the original 360 Hz.

INT8 computation. The CNN accelerator on the MAX78002 works with low-precision integers (we use 8-bit) instead of 32-bit floating-point numbers, so the model must be quantized. Clipping the input to ± 4 keeps the value range small, and if quantization costs too much performance, we can try quantization-aware training.

Class imbalance. In our prepared dataset N makes up 89.5% of all beats, V 7.0%, S 2.8% and F only 0.8%, so the model sees very few examples of the rare classes. We plan to use a class-weighted loss for this (Section 3.6).

Few subjects. The database has 48 recordings from only 47 people (Moody and Mark, 2001), and after removing the paced records just 22 recordings are left for training and validation. Records 201 (DS1) and 202 (DS2) come from the same person, but since 201 is in our validation set, the model never trains on that person.

Noise. MIT-BIH recordings are fairly clean compared with what a wearable records in daily life, where baseline wander, muscle noise and electrode motion are common. We filter the signal

(Section 3.4) and test the model with added real noise (Section 5.4).

Real-time use. A device working on a live signal only has past samples, so our filters are causal (Section 3.4). Because the input includes the time until the next beat, each beat can only be classified once the next beat has arrived. In our data the median gap between beats is 0.74 s, so this delay is small enough for monitoring.

Computational resources. In a course project the time for training and trying out models is limited, so we keep the models small and the number of experiments manageable.

No MAX78002 board. Without the hardware we cannot measure inference time, power use or the real accuracy on the chip, so we can only verify the model in software (Section 6.4).

1.6 Risks and Challenges

These risks follow directly from the data-preparation work in Section 3, each one is something we measured in this dataset, not a generic list.

Table 2: Risks, their importance and proposed mitigations

Risk	Why it is important	Proposed mitigation
Severe, patient-concentrated class imbalance. F is 0.8 % of all beats overall, and one record supplies ∼90–95 % of it in both the development set (208) and the test set (213) (Sections 3.6, 3.8).	A model can reach high overall accuracy while rarely predicting F or S correctly. Because F is concentrated in one or two patients, good F performance may reflect the model learning one person's ECG rather than the fusion-beat class in general, and would not be visible if we only looked at accuracy.	Report macro-F1 and per-class precision/recall/F1 rather than accuracy (Section 5); use a class-weighted loss set from the training split only (Section 3.6); treat validation-F (only 21 beats, Section 3.1) as a coarse pass/fail alarm rather than a metric to optimise; run patient-level error analysis on DS2 (Section 5.5).
S beats look like normal beats and are only separable by when they arrive.	S beats are premature beats that start above the ventricles, so their waveform is almost the same as a normal beat. The main cue is that they come too early. A model that relies on the waveform alone will mistake S for N, and because N is 89.5 % of all beats while S is only 2.8 % (Section 1.5), these errors barely change accuracy but cost S recall. The timing cue is also fragile: it depends on the neighbouring R-peaks being detected correctly, and it is undefined for the first and last beat of a record, which are dropped.	Give the model the three RR intervals (previous, next and the median of the last four gaps) together with the beat window (Section 1.2), so that an early beat can be compared with the patient's recent rhythm. Compare waveform-only against waveform + RR to show that the timing information is what separates S from N. Report per-class S recall and the S to N confusion rather than accuracy (Section 5), use the class-weighted loss (Section 3.6), and include S beats in the patient-level error analysis on DS2 (Section 5.5).

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Table 2 – continued from previous page

Risk	Why it is important	Proposed mitigation
Data leakage from patient-correlated morphology if beats were split randomly rather than by record.	Consecutive beats from the same patient share electrode placement, baseline noise and individual anatomy. A random beat-level split would let the model see near-duplicate waveforms in both training and test, inflating reported performance far above what a genuinely unseen patient would give.	Strict inter-patient DS1/DS2 partition, with the internal train/validation split inside DS1 also performed by record, never by beat (Section 3.1). DS2 stays untouched until final evaluation; no hyperparameter tuning is done against it.
FP32 → INT8 quantisation may hurt minority classes more than the aggregate accuracy figure suggests.	Quantisation error behaves like added noise. Classes with narrower decision margins (S, F) are more likely to flip near a quantisation boundary than the majority N class, so a small aggregate accuracy drop could hide a much larger relative loss on the classes this project cares about clinically.	Report FP32 vs INT8 per-class metrics side by side, not only the aggregate (Sections 5.1–5.3); keep quantisation-aware training available if the drop is large; constrain the architecture early to operations the MAX78002 accelerator supports, leaving headroom before synthesis is attempted.
Gap between MIT-BIH recording conditions and real ambulatory wearable use.	MIT-BIH recordings are calmer than a wearable will encounter in daily use. A model that scores well on clean MIT-BIH beats could still fail in the field, and reporting only clean-test numbers would overstate real-world readiness.	Explicit noise-robustness protocol (clean / 12 dB / 6 dB / 0 dB) using disjoint train/test halves and channels of the MIT-BIH Noise Stress Test Database, so the robustness numbers cannot be leaked or memorised (Section 3.7).
No physical MAX78002 development kit is available for this project.	Without hardware in hand, real inference latency, power/energy draw and true on-device INT8 accuracy cannot be measured. Treating a successful software toolchain pass as equivalent to a hardware-verified result would overstate deployment readiness.	Keep a clear PASS/FAIL software-verification table (toolchain acceptance, quantisation, synthesis, generated files) separate from any hardware claim (Sections 6.3–6.4), and document what could not be tested with a proposed hardware validation plan rather than estimating numbers we cannot measure.

2 Data Understanding

2.1 Dataset Description

We use the MIT-BIH Arrhythmia Database (Moody and Mark, 2005), version 1.0.0, available from PhysioNet (Pollard et al., 2026). The complete database contains 48 approximately half-hour ECG recordings from 47 people. Each recording contains two ECG channels sampled at 360 Hz.

The .dat files contain the ECG measurements, .hea files describe recording properties, and .atr files contain the reference annotations. Sampling frequency, channel names and recording duration were obtained from the DS1 headers. Class counts were calculated from the DS1 annotation files.

Table 3: Description of the complete source database and the DS1 subset

Parameter	Complete Source Database	DS1 Analysis
Database Name	MIT-BIH Arrhythmia Database, version 1.0.0	DS 1 Subset
Number of records	48	22
Number of subjects	47	22
Number of channels	2 per record	2 (44 channel traces examined)
Sampling Frequency	360 Hz per channel	360 Hz per channel
Recording duration	30.093 minutes	30.093 minutes per recording
Annotation type	Expert-reviewed beat, rhythm, signal-quality and other event annotations in .atr files	Same annotations, restricted to DS1
ECG Leads available	MLII, V1, V2, V4, V5	MLII, V1, V4 and V5 observed

The analysed DS1 records were:

101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 201, 203, 205, 207, 208, 209, 215, 220, 223 and 230.

The supplied DS1/DS2 membership was retained. All 48 recordings were checked for file availability, while waveform and annotation analysis was restricted to DS1. The results below describe raw data before model preparation.

2.2 Target Class

Table 4: Target classes in DS1: original annotations, number of beats and percentage

Final Class	Original Annotations included	Number of beats	Percentage (%)
N	"N, L, R"	45834	89.85
S	"A, a, J, S, e, j"	976	1.91
V	"V, E"	3788	7.425
F	"F"	415	0.813

Annotations outside these four classes were excluded from target-class counts. Eight unclassifiable beats marked Q were excluded because they have no reliable label among the required classes. A further 1,515 annotations described other events rather than N/S/V/F heartbeat targets. These remained available for rhythm and quality inspection.

The protocol excludes paced recordings 102, 104, 107 and 217. Paced-beat symbols / and f also fall outside the selected mapping. So, neither was observed in the analysed DS1 annotations. These are class-definition exclusions.

Table 5: Annotations outside the four target classes in DS1

Symbol	Category	Meaning	Count
+	Non-beat annotation	Rhythm change	616
!	Non-beat annotation	Ventricular flutter wave	472
∼	Non-beat annotation	Signal-quality change	289
|	Non-beat annotation	Isolated QRS-like artifact	66
x	Non-beat annotation	Blocked P-wave	58
Q	Excluded beat	Unclassifiable beat	8
[	Non-beat annotation	Start of ventricular flutter/fibrillation	6
]	Non-beat annotation	End of ventricular flutter/fibrillation	6
“	Non-beat annotation	Comment	2

2.3 Exploratory Data Analysis

2.3.1 ECG Signal and Noise Example

Figure 1 shows reference-annotated clean, noisy and unreadable MLII intervals from recordings 101, 108 and 208. The clean interval has clearly distinguishable repeating beats. The noisy interval has an unstable baseline and additional fluctuations, while the unreadable interval shows substantial drift and poorly distinguishable beats. More examples of noisy beats are provided in Figure 7

2.3.2 Example Heartbeat from each class

Figure 2 shows one annotated example from each target class: N and S from recording 101, V from 106, and F from 108. MLII was selected by its header name to provide a consistent measurement view. The distribution of the four classes can be visualized in **Figure 3**.

2.3.3 Subject/Record Distribution

Figure 4 shows different numbers of retained beats across the approximately equal-duration recordings. Counts range from 1,619 in recording 124 to 3,363 in recording 215.

The record and subject distributions match within DS1 because each analysed recording belong to a different person. Subject key 201_202 identifies the person whose second recording lies in DS2 and only record 201 contributes to these DS1 statistics.

Figure 5 shows that class availability also varies between recordings.

2.3.4 Signal Amplitude Distribution

Figure 6 displays amplitude histograms for the observed leads. We sampled 5,000 values from each record/channel using random seed 42, giving 220,000 plotted measurements. Histogram widths were calculated using the data-dependent Freedman–Diaconis rule.

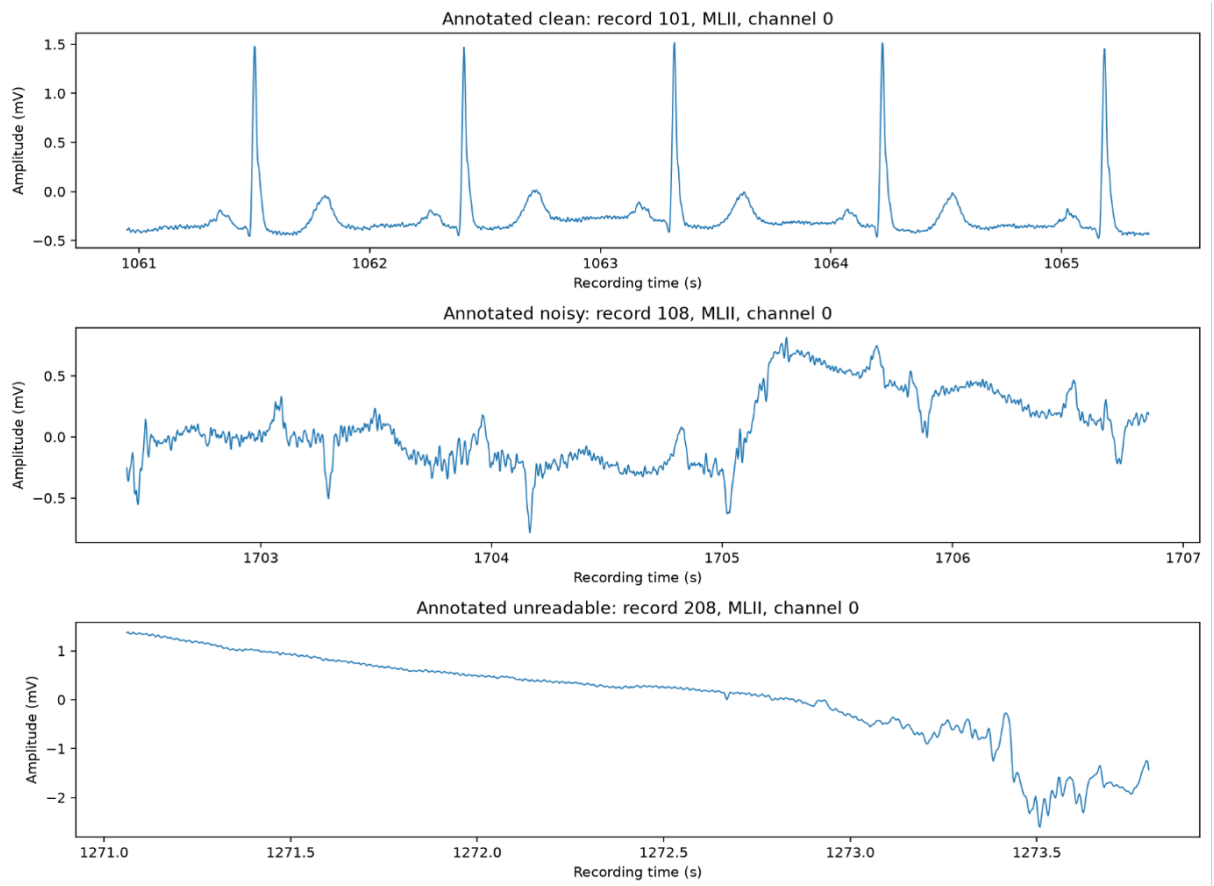


Figure 1: Different types of MLII intervals. Top panel: reference-annotated clean interval. Middle panel: noisy interval. Bottom panel: unreadable interval.

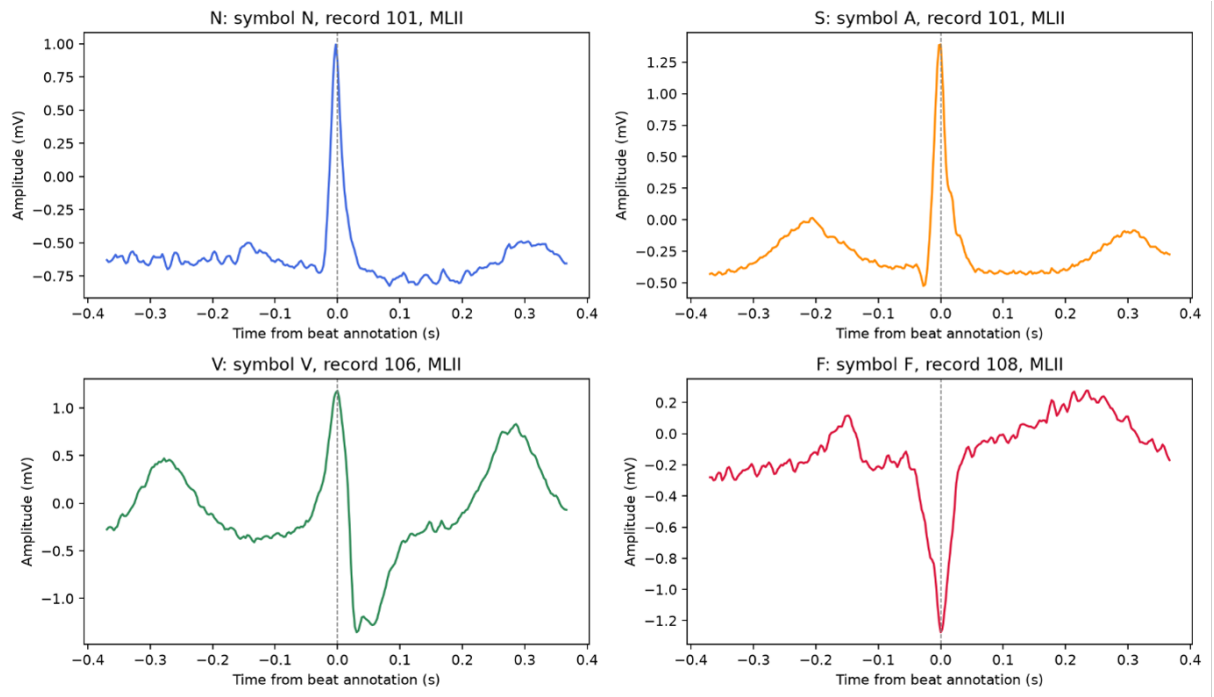


Figure 2: Heartbeat classes. Top left: N, Top right: S. Bottom left: V. Bottom right: F.

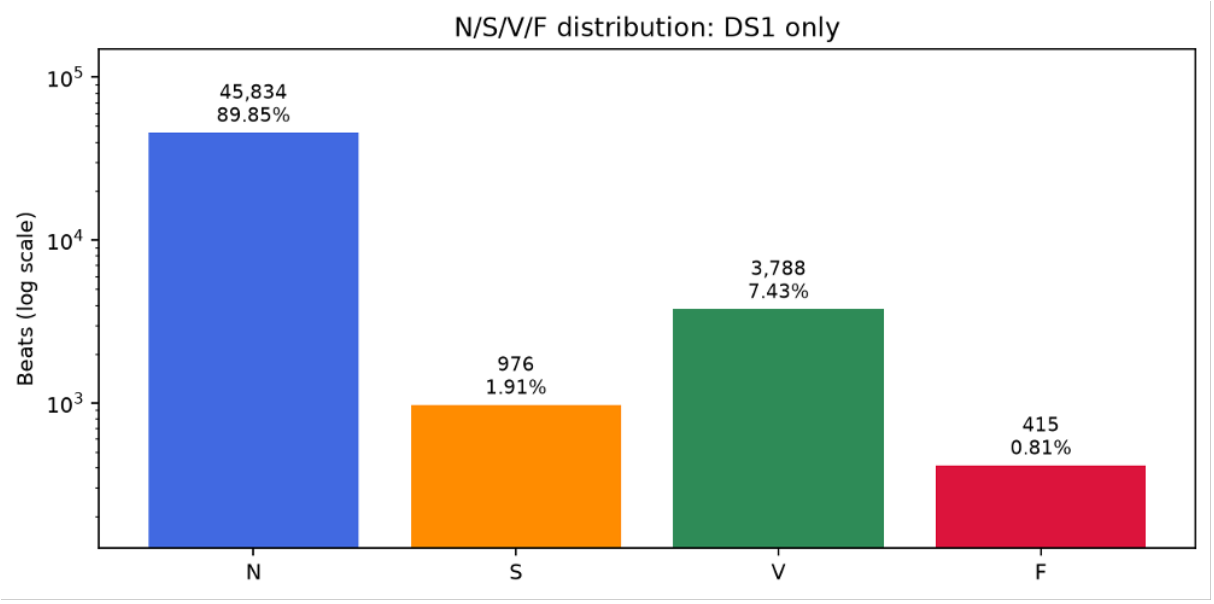


Figure 3: Heartbeat classes distribution.

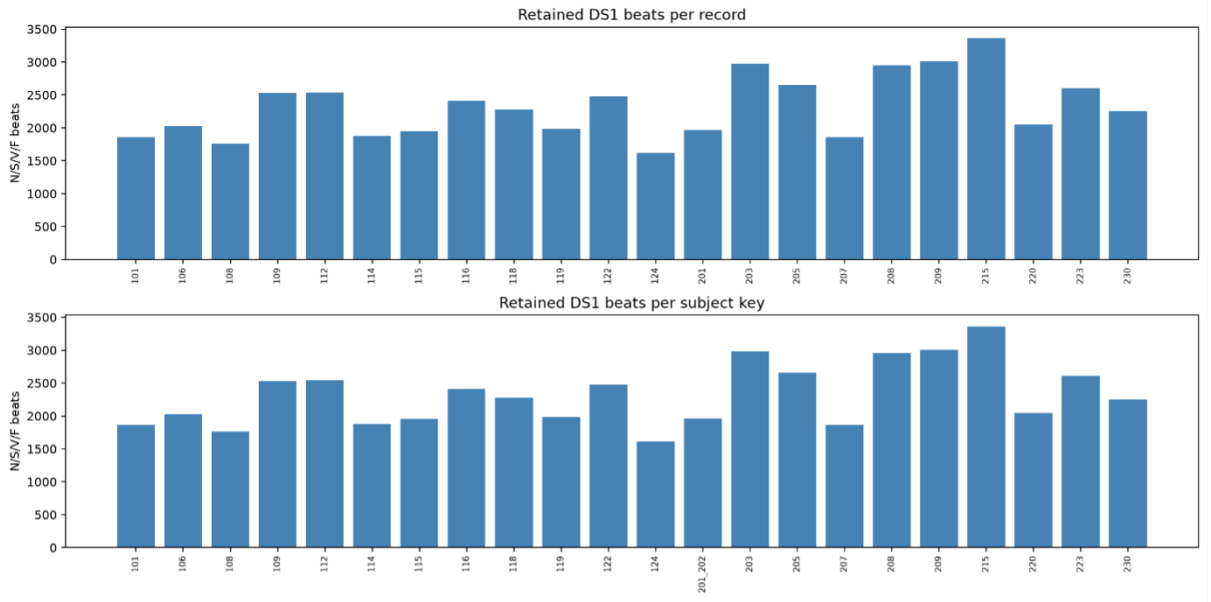


Figure 4: Number of retained beats per recording. Numbers of retained beats across the approximately equal-duration recordings. Top panel: retained beats per record. Bottom panel: retained beats per class

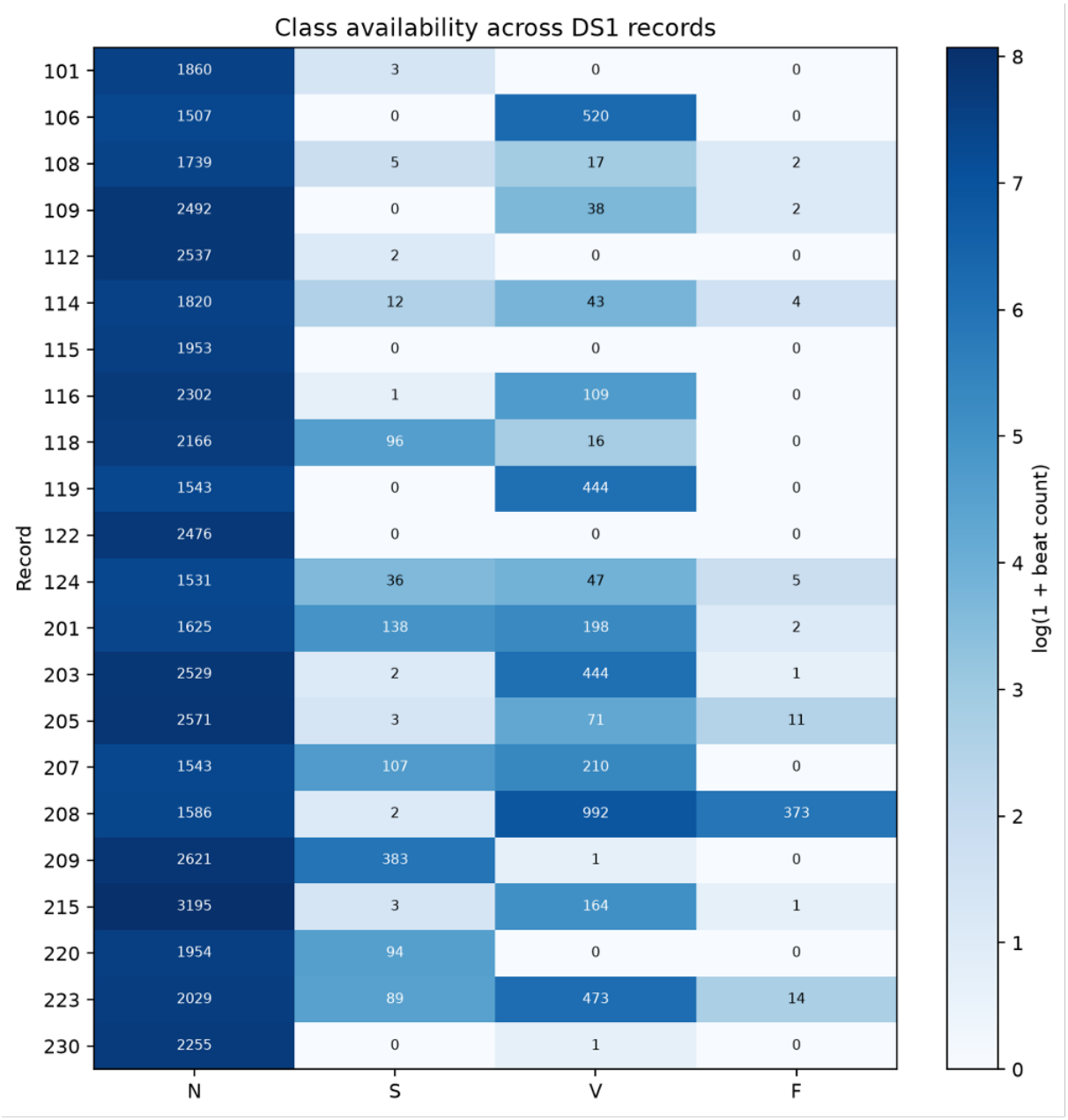


Figure 5: Class availability per recording. Variation in class availability between recordings

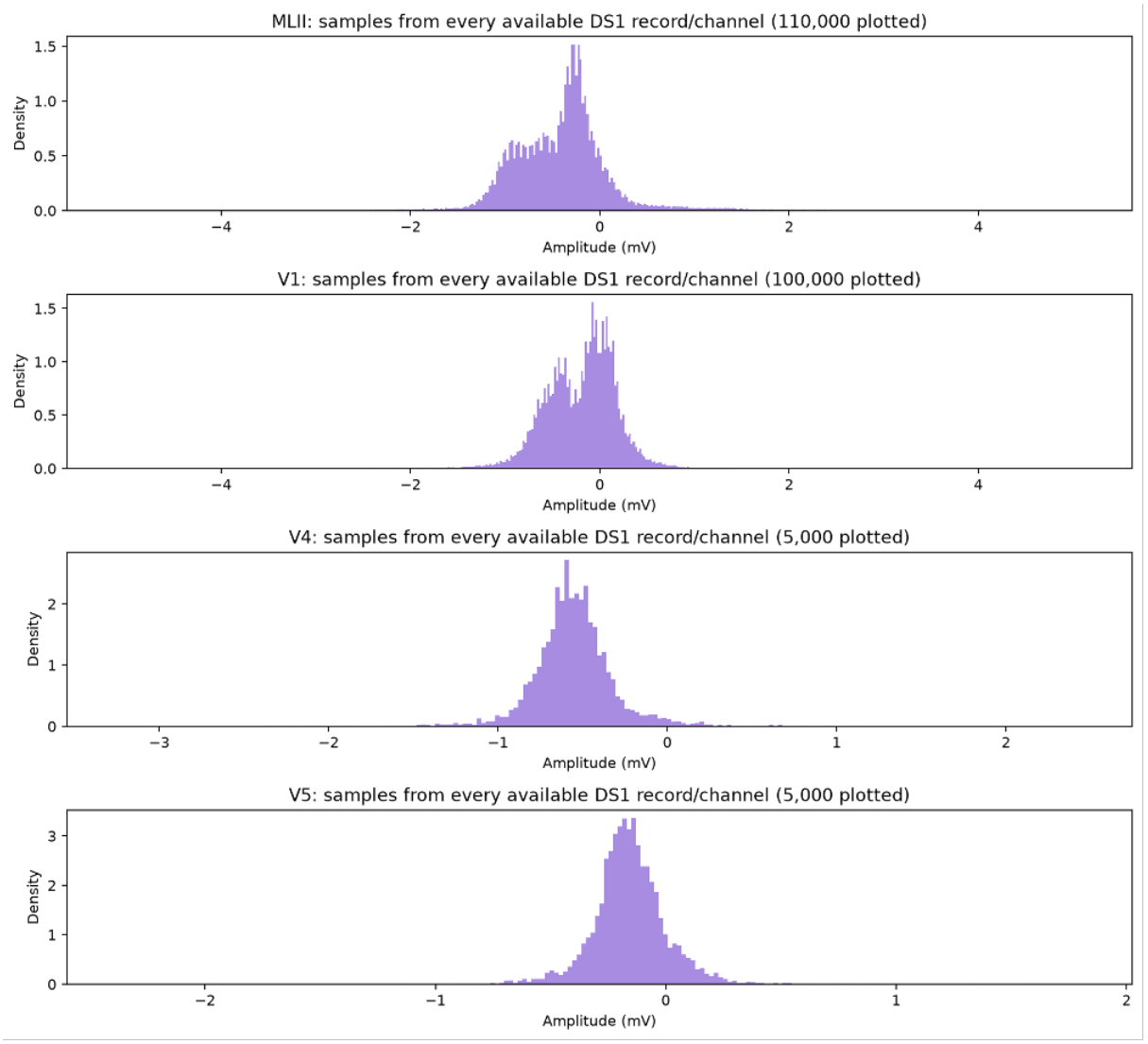


Figure 6: Amplitude histograms for the observed leads. From top to bottom: MLII, V1, V4 and V5

2.3.5 Examples of Noisy ECG

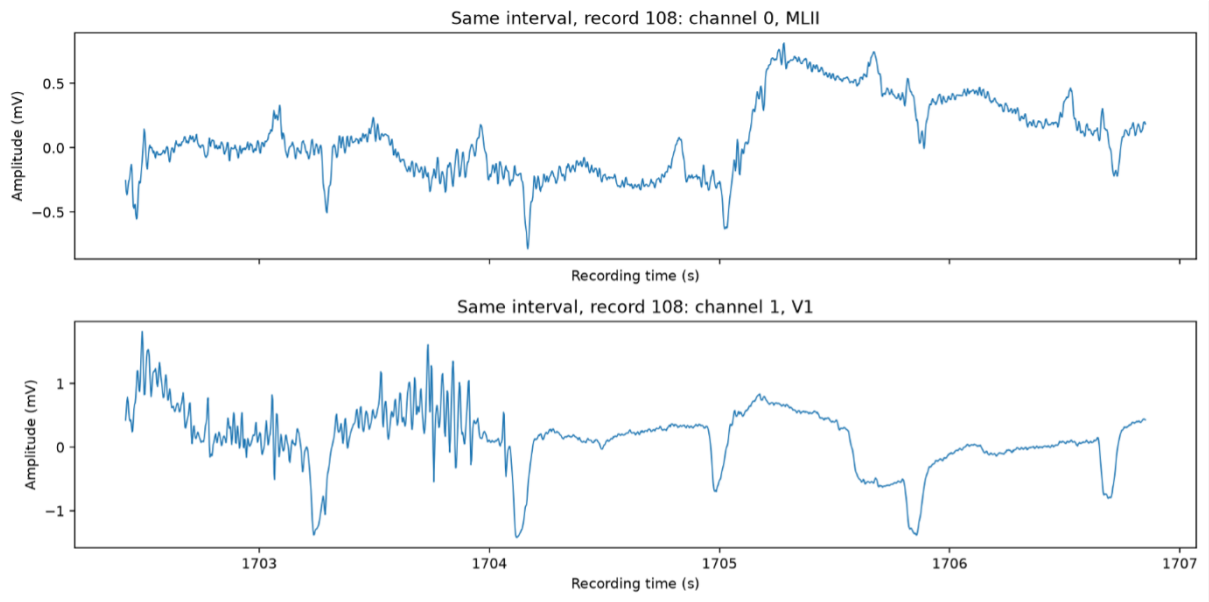


Figure 7: Examples of noisy ECG.

2.4 Data Quality

Table 6: Data quality checks on the DS1 recordings

Quality Check	Result
Successfully read DS1 recordings	22
Channel traces examined	44
Signal read errors	0
Missing values represented by NaN	0
Infinite sample values	0
Header checksum mismatch	0
Samples at or beyond computed ADC limits	1635
Out of range annotation positions (DS1)	0
Out of order annotations (DS1)	0
Duplicate beat positions (DS1)	0

Extreme Amplitudes and potentially corrupt segments: Observed amplitudes ranged from −5.120 to 5.115 mV. These extrema occurred in MLII from recording 116 and V1 from recording 223.

The 1,635 samples flagged at possible ADC limits identify candidate clipping regions. No channel had an exact constant digital run lasting at least 0.7389 seconds. Large jumps and constant digital runs were also measured. Recording-limit values, sudden changes and repeated samples are quality indicators, and do not independently prove signal corruption. No automatic deletion was based on these checks.

Class Imbalance: 'N' represents 89.85% of retained DS1 beats. A rule that assigns every retained DS1 beat to N would therefore match 89.85% of the labels while identifying none of the minority classes. This supports the use of macro-F1 and per-class recall described in Section 1. The concentration of F beats in recording 208 additionally limits the range of subjects represented by that class.

Difference between recordings and channels: Figure 8 compares median amplitudes, the middle 50% of sample values and annotated noisy/unreadable duration for each channel. Median offsets, amplitude spread and quality coverage vary between recordings and between their two channels.

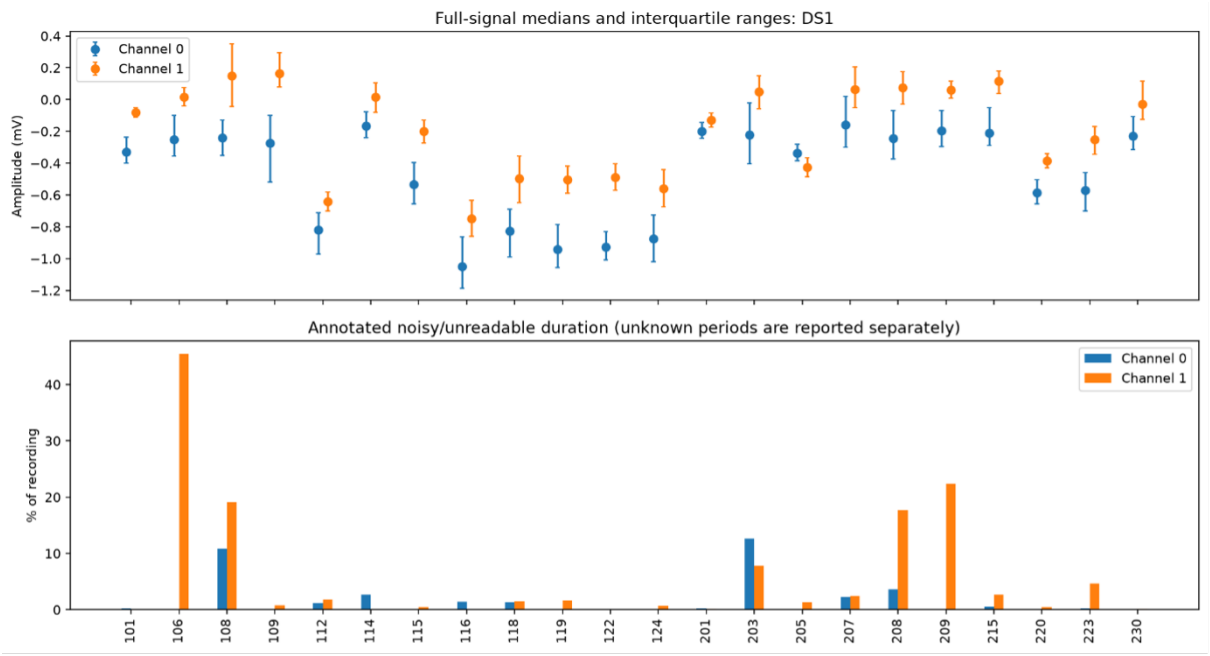


Figure 8: Median amplitude, interquartile range and annotated noisy/unreadable fraction per recording and channel.

For example, recording 106 has a much larger annotated noisy/unreadable fraction in channel 1 than in channel 0.

2.5 Leakage Analysis

Randomly splitting individual heartbeats from the same person between training and testing can produce misleading results. These beats share characteristics of that person's ECG and recording conditions. Neighbouring beats can also be highly similar, and extracted windows may share signal samples. A model can therefore exploit familiar patterns in the test group, making its performance appear better than its performance on previously unseen people. Multiple beats from one person are not equivalent to the same number of independent people.

Our evaluation design follows the supplied whole-record DS1/DS2 partition. DS1 is used for development, and DS2 is reserved for final testing. All heartbeats from a recording belong to its assigned group instead of being randomly distributed between development and testing. This prevents the same recording from supplying beats to both groups. The development design also separates training and validation by recording, the EDA notebook itself does not create these model-training splits.

Leakage could also arise from using test data to select model or preprocessing settings, fitting normalisation on combined training and test data, or distributing original and augmented versions of a beat across groups. The EDA does not perform fitted normalisation, training augmentation or model training.

The supplied DS1/DS2 partition nevertheless contains one documented subject overlap: recordings 201 and 202 belong to the same person, with 201 assigned to DS1 and 202 to DS2. A model trained on 201 and evaluated on 202 would therefore encounter a familiar person despite receiving a different recording. The supplied partition separates recordings but is not fully subject-independent. Our current analysis identifies this limitation, restricting access to DS2 alone does not eliminate it.

3 Data Preparation

3.1 Record Selection

We follow the standard inter-patient partition (de Chazal et al., 2004) required by the project rather than a random beat-level split. Splitting by record guarantees that no validation or test beat comes from a patient whose beats also appear in training.

DS1 – development set (22 records), used for training and validation: 101, 106, 108, 109, 112, 114, 115, 116, 118, 119, 122, 124, 201, 203, 205, 207, 208, 209, 215, 220, 223, 230.

DS2 – final test set (22 records), never used for training, validation or model selection: 100, 103, 105, 111, 113, 117, 121, 123, 200, 202, 210, 212, 213, 214, 219, 221, 222, 228, 231, 232, 233, 234.

Records 102, 104, 107 and 217 do not appear in either list. All four are paced recordings, which the project permits us to exclude, de Chazal's original partition excludes them for the same clinical reason, and keeping them out preserves comparability with the literature this partition comes from.

Within DS1 we set aside five records as an internal validation set, again split by record rather than by beat, per `configs/prep.yaml`: 106, 112, 124, 201, 223. The remaining 17 DS1 records form the training set: 101, 108, 109, 114, 115, 116, 118, 119, 122, 203, 205, 207, 208, 209, 215, 220, 230.

This split is not accidental for the minority classes. Between them, the five validation records hold 234 of DS1's 944 S beats (25 %) and 21 of its 415 F beats (5 %), so validation can detect a model that fails on S or F outright. The F-heavy record 208 (372 F beats, 90 % of DS1's F beats) stays in training rather than validation, since its beats are far more useful for the network to learn from than to be held back for a validation set they would otherwise dominate. Section 3.8 returns to the consequence: with only 21 F beats, validation F is read as a coarse alarm rather than a trustworthy estimate.

3.2 ECG Channel Selection

We use one channel throughout the modified limb lead MLII. Three things decided it. It appears in all 44 records of the inter-patient partition, so we lose no patient. Its QRS Complex is large and mostly positive, which is makes the R-Peak the clearest landmark in the signal and every window in Section 3.3 is placed relative to that pea. And it is the channel used in the literature that defines this partition, so our numbers stay comparable with published ones. Sticking to one channel also halves preprocessing cost and input memory on the device. Which matters here a single-lead wearable is what this model would actually run on.

We pick the lead by name from the record header, never by channel number. This matters more than it looks. Channel order is not consistent across the database: record 114 stores its channels as V5 | MLII, the reverse of the usual arrangement. Had we simply read channel 0, that one patient would have been trained on V5 while the other 43 supplied MLII — an error that produces no warning and would be very hard to find later. If MLII were missing we fall back through V5, V2, V1, V4 and log the substitution, though across our 44 records the fallback never fired (`results/lead_audit.csv`). Records 102 and 104 carry no MLII at all, but both are paced and already excluded on clinical grounds.

3.3 Beat Extraction

We take R-peak positions from the reference .atr annotations that ship with the database, placed and verified by cardiologists. We do not run an automatic QRS detector. As a real wearable has no cardiologist, and its own detector would make mistakes that feed straight into both the window position and the RR intervals we derive from it. Before any of this we strip the non-beat markers — rhythm (+), signal quality (∼) and artefact (|). There are 2,991 of them across the 44 records, and treating them as heartbeats would inflate every class total we report.

Each window takes 48 samples before the R-peak (266.7 ms) and 80 after it (444.4 ms), 128 samples in total, covering 711 ms at our working rate of 180 Hz. The asymmetry is deliberate. Ahead of the R-peak the only thing we need is the P wave, which sits roughly 200 ms earlier, while repolarisation behind it runs on much longer. A symmetric window of the same length would have to either cut off the T wave or waste samples on flat baseline. Choosing 128 also keeps every intermediate tensor inside the per-layer dimension limits of the MAX78002.

One complication follows from filtering causally: it delays the signal, so the annotated positions no longer line up with the waveform we have processed. We measured the delay rather than calculating it, which turned out to matter our own filter chain accounts for only about 3 samples, and the rest comes from the anti-aliasing filter hidden inside the decimation step. The total is 8 samples at 360 Hz, or 22.2 ms. We shift the annotations by that amount before cutting. Left uncorrected the R-peak settles at index 52 rather than 48; corrected, it sits at 48. We checked this across twelve records (`scripts/verify_alignment.py`).

We drop a beat if it sits too close to the start or end of a recording, or if it has no neighbour on one side and so no measurable RR interval. In practice only the second case happens, the first and last beat of each record, 88 altogether, or 0.087 % of the data.

3.4 Signal Preprocessing

We apply the following operations in this order.

Table 7: Signal preprocessing operations, applied in order

#	Operation	Parameters	Why it is necessary
1	Lead selection	MLII, by name	Same physical lead for every patient (Section 3.2)
2	Causal filtering	Butterworth order 3; high-pass 0.5 Hz, notch 60 Hz, low-pass 40 Hz	Removes baseline wander, mains interference and muscle noise, none of which carry cardiac information.

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Table 7 – continued from previous page

#	Operation	Parameters	Why it is necessary
3	Decimation	360 to 180 Hz, anti-aliased	Halves input length and on-device data memory; content above 40 Hz is already removed, so nothing diagnostic is lost.
4	Normalisation	Per record: subtract median, divide by 99.5th percentile of absolute deviation	Removes per-patient sensor gain while preserving relative amplitude within a patient, which is a real cue.
5	Clipping	±4	Bounds the range so rare artefacts cannot dominate the INT8 quantisation scale.
6	Delay compensation	+8 samples, measured	Realigns annotations with the filtered signal (Section 3.3).
7	Segmentation	48 before, 80 after, 128 total	Produces the fixed-size network input (Section 3.3).

- Causality. Our filters run in one direction only. We could have used zero-phase filtering most of the ECG literature does but it passes over the signal forwards and then backwards, which means it needs samples from the future. A device classifying a live signal does not have them. Reporting results obtained that way would mean reporting something the hardware could never reproduce. The price we pay is the group delay measured in Section 3.3.

- Cutoffs. The 0.5 Hz high-pass clears the slow drift caused by breathing, while staying low enough to leave the ST segment alone. The notch sits at 60 Hz rather than 50 Hz because these recordings were made in the United States. The 40 Hz low-pass removes muscle artefact and conveniently doubles as the anti-aliasing filter for step 3.

- Decimation, not simply dropping samples. Keeping every second sample would fold any content above 90 Hz back into our signal band, and that cannot be undone afterwards. We stop at 180 Hz rather than going lower because QRS width is what separates normal beats from ventricular ones, and decimating too far blunts exactly that.

- Choice of scale. We normalise per record rather than per beat, because how much taller one beat is than its neighbours is itself a clue and per-beat scaling would erase it. Our first attempt used the median absolute deviation and had to be abandoned: the flat baseline dominates that statistic, so R-peaks ended up anywhere between 6 and 23 scale units depending on the patient, and any clip low enough to be useful would have sliced through the QRS. Scaling by the 99.5th percentile instead puts them between 0.46 and 1.01 on every record, and the clip at 4 is never reached the largest value across all 100,630 beats is 2.25.

- RR features. Besides the waveform we store three timing values for every beat: `rr_prev`, the gap to the previous beat; `rr_next`, the gap to the next; and `rr_local`, the median of the previous four gaps, which stands for the patient's current rhythm. All three are in seconds and are computed from the annotation positions at 360 Hz, before resampling, so decimation cannot shift them. They matter because an S beat is defined by arriving early rather than by looking different, and a 128-sample window centred on a single beat cannot express that. One detail decides whether they are correct: the gaps are measured with every annotated beat still present, including the paced and unclassifiable beats that

are excluded later. Removing a beat first would merge two gaps into one and invent an interval twice as long as anything the heart produced. All three values are stored per beat alongside the waveform in `data/processed/beats.npz`.

3.5 Label Preparation

Beat-level annotations use the raw MIT-BIH symbol set. Section 3.3 already removes annotations that are not beats at all - rhythm (+), signal-quality (∼) and artefact (|) markers, 2,991 of them across the 44 records. What remains is mapped onto the four AAMI EC57 classes required by the brief, following the same grouping used by de Chazal et al. (2004):

Table 8: Mapping from MIT-BIH annotation symbols to the four target classes

Final class	Original MIT-BIH symbol(s)	AAMI EC57 rationale	Beats (44 records)
N	N, L, R, e, j	Normal beats plus bundle-branch-block conduction (L, R) and escape beats (e, j): morphologically distinct but not ectopic	90,125
S	A, a, J, S	Supraventricular ectopic beats: atrial/nodal premature and aberrated beats	2,781
V	V, E	Ventricular ectopic and escape beats	7,009
F	F	Fusion of a ventricular and a normal beat	803
Excluded (Q)	/, f, Q	Paced beats and beats the original annotators could not classify; not part of the 4-class task	15

`src/io_utils.py` implements this mapping as a single dictionary (`AAMI_MAP`) rather than scattering it through the pipeline, so the class definition has one place to audit. `src/data_audit.py` checks every beat symbol that appears in the 44 records against it; the result (`results/audit_summary.txt`) is that none fall outside the table above - there is no 'unmapped' bucket left over. The 15 excluded Q beats are counted, not silently dropped, so the exclusion is documented rather than assumed.

A further 88 beats (0.087 % of the 100,718 N/S/V/F beats) are dropped later, in Section 3.3, because they are the first or last beat of a recording and so have no measurable RR interval, that drop happens after labelling and is independent of it (Section 3.8).

3.6 Class Imbalance

Section 3.8 reports the full split table and the headline numbers are that F is 0.8 % of the whole dataset and S is 2.8 %, against 89.5 % for N. This is heartbeat-database imbalance, not an artefact of our pipeline, de Chazal's own inter-patient partition has the same shape, and the alternative (a balanced sample) would no longer be representative of what an ambulatory monitor actually records.

The imbalance is also concentrated by patient rather than spread out: one record (208 in development, 213 in test) supplies roughly 90–95 % of that split's F beats (Section 3.8), and the test split's S share (3.70 %) does not match the training split's (1.76 %) because the published DS1/DS2 partition was fixed by record, not by class balance.

We do not correct this at the data-preparation stage. Three options were considered and rejected for the reasons already given in Section 3.7: SMOTE-style interpolation invents waveforms for a class held by one or two patients; random undersampling of N would shrink an already small (100k-beat) dataset and is unnecessary since the model will be judged on macro-F1 and per-class metrics rather than accuracy; and duplicating minority beats verbatim would let the network memorise a handful of individual waveforms rather than learn the class. What we defer to the Modeling stage (Submission 2, Section 4) is: (i) a class-weighted loss, with weights set from the training split's inverse class frequency only, so DS2 and the validation records never influence it; (ii) evaluating with macro-F1 and per-class precision/recall/F1 rather than accuracy, so a model that only predicts N is not rewarded; and (iii) treating validation-F, which has only 21 beats (Section 3.1), as a coarse pass/fail alarm rather than a metric to optimise against. The augmentation policy in Section 3.7 is applied per beat regardless of class, so it increases how many distinct-looking versions of a minority beat the network sees per epoch without inventing new ones - a mild, indirect form of oversampling we consider acceptable where SMOTE is not. We also considered balanced-batch sampling and focal loss as alternatives to a plain class-weighted loss; both stay on the table for model selection in Submission 2 (Section 4.4), compared against class weighting on the same validation split.

3.7 Data Augmentation

We augment the training split only. Validation and test data are left untouched — augmenting them would quietly change the thing we are trying to measure.

Table 9: Data augmentation transformations

Transformation	**Range**	**Why it is realistic**
Amplitude scaling	$\times 0.9$ to $\times 1.1$	Variation in electrode contact and sensor gain between wearing sessions.
Temporal jitter	± 4 samples	The localisation error an on-device QRS detector makes against an expert annotation.
Additive real ECG noise	SNR 6 to 20 dB	The noise actually present in ambulatory recording.

The noise comes from the MIT-BIH Noise Stress Test Database, which provides baseline wander, muscle artefact and electrode motion as separate recordings. We use real noise rather than synthetic Gaussian noise because ECG noise has structure: baseline wander is slow drift, electrode motion is a series of abrupt jumps. Gaussian noise looks like neither, and a model trained on it would be ready for neither.

The noise we train on and the noise we test on must not overlap. We split each noise recording so training draws one half and one channel while evaluation gets the other. Skip this and a model can simply memorise the particular noise waveform it saw, which makes the resulting robustness figure worthless. For the same reason we do not evaluate on the twelve ready-made noisy records that come with the database: all twelve are built from records 118 and 119, and both are training records in our partition.

We deliberately leave out three things. Time warping, because it changes the RR relationships the S class is defined by. Synthetic minority oversampling, because interpolating between beats of a class held by one or two patients invents waveforms that belong to nobody. And reflection or sign inversion, because the result is not an ECG any heart could produce.

3.8 Final Dataset

Table 10: Final dataset: number of records and beats per class in each split

Split	Records	N	S	V	F	Total	F%
Train	17	36574	709	2550	393	40226	0.98
Validation	5	9250	234	1238	21	10743	0.20
Test (DS2)	22	44218	1836	3219	388	49661	0.78
Total	44	90042	2779	7007	802	100630	0.80

These totals come directly from the run described in Section 3.9 (`scripts/run_prepare.py`, 0 reconciliation mismatches) and match the record lists fixed in Section 3.1.

Of the 100,718 beats belonging to our four target classes we extracted 100,630 and dropped 88, which is 0.087 %. All 88 are the first and last beat of a recording; none was lost to a window running off the end. A further 15 had already been excluded as paced or unclassifiable. Every beat keeps its record number, so the data can be re-split later without running preprocessing again.

- Validation holds just 21 F beats. That is enough to notice if the model fails on F completely, but nowhere near enough for an F1 score we would trust, so we plan to select models on the other three classes and read validation F only as a rough alarm.
- Class F is not simply rare, it is concentrated in one person. Record 208 supplies 372 of the development set's 414 F beats; the next largest contributor supplies 14. Whatever we learn about F, we largely learn from one patient.
- The test split is not composed like the development split, S makes up 3.70 % of it against 1.76 % of training. That is a property of the published partition rather than a mistake on our side, and it is one more reason we report per-class metrics instead of accuracy.

3.9 Reproducibility

Everything needed to regenerate the dataset from scratch is pinned in `requirements.txt` and `configs/prep.yaml`, so the process does not depend on remembering versions or parameters. We ran the pipeline under Python 3.14.7 with PyYAML 6.0.3, NumPy 2.5.2, pandas 3.0.5, SciPy 1.18.1 and wfdb 4.3.1 (matplotlib 3.11.1 and torch 2.14.0 are pinned for later stages but unused by data preparation).

A single seed, 20260924 (`configs/prep.yaml` → `reproducibility.seed`), is defined for the whole project. Data preparation itself is deterministic - filtering, decimation, normalisation, segmentation and the AAMI mapping involve no randomness, so the seed does not change any number in this report; it exists so that the data augmentation (Section 3.7) and model training (Submission 2) that consume this dataset are themselves reproducible.

Every parameter used in Sections 3.2–3.8, sampling rate (360 → 180 Hz), filter cutoffs and order, normalisation percentile and clip, the +8-sample delay compensation, the 48/80-sample window and the record lists themselves is read from `configs/prep.yaml` at run time rather than hard-coded (Section 3.4), so this file is the single source of truth for reproducing the exact dataset reported here. The network input this produces is a single fixed-size vector: 128 samples (48 pre-R, 80 post-R) from one lead (MLII) at 180 Hz, per beat (Section 3.3).

To reproduce, create the environment from `requirements.txt`, then run `scripts/run_prepare.py` from the project root (or `./run.sh`, which does the same). The script downloads the 44 records once via `src/io_utils.py`'s `ensure_database()` (`wfdb.dl_database`) into `data/mitdb/`, marked complete with a `.complete` file, so repeated runs read from that local cache instead of streaming from PhysioNet every time. We do not submit that folder, the README explains how to regenerate it locally.

The run is checked, not just executed: `src/data_audit.py` cross-checks every mapped beat against the published de Chazal counts before preparation (`results/audit_summary.txt` - 0 mismatches, 0 unmapped symbols), `scripts/run_prepare.py` reconciles kept + dropped beats against that audit for all 44 records after preparation (0 mismatches, Section 3.8), and `scripts/verify_alignment.py` independently checks the R-peak delay compensation from Section 3.3 on twelve records. All three passed on the run this report is based on (`results/dataset_stats.csv`, `results/split_stats.csv`).

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